

# VANGALA JAYA SAI KUMAR REDDY

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LinkedIn • GitHub

## Career Objective

Computer Science student seeking opportunities in software development and machine learning to build scalable, real-world applications and continuously enhance technical and problem-solving skills.

## Skills

**Languages:** Java, Python, C, SQL, JavaScript

**Web Technologies:** HTML, CSS, React, Node.js

**Tools & Platforms:** Git, Postman

**Core Concepts:** Data Structures and Algorithms, Machine Learning, System Design, Data Analytics

## Projects

### Digital Twin-Based Bus Transportation System

Dec 2025 - Feb 2026

GitHub

- Designed and developed a multi-layer Digital Twin architecture to model buses, routes, bookings, and trip operations with synchronized digital representations.
- Implemented backend validation and transactional logic to handle concurrent booking requests, preventing seat over-allocation and ensuring data consistency.
- Executed 25+ operational test scenarios validating system behavior across booking, cancellation, and trip lifecycle workflows.
- Achieved real-time synchronization between frontend, backend, and database layers, ensuring consistent system state across all entities.

### EcoLink – Recyclable Waste Collection Platform

Oct 2025-Dec 2025

GitHub

- Built a full-stack platform enabling users to schedule recyclable waste pickups with flexible reward options (money, essentials, donation).
- Designed a zone-based threshold system to optimize collection routes, reducing unnecessary trips by approximately 30% during testing.
- Implemented real-time status updates across user and recycler dashboards, improving transparency in pickup scheduling and completion.
- Developed backend logic using PHP and MySQL to manage user data, appointments, and dynamic waste aggregation across zones.

### Personal Portfolio Website

May 2025

Live

- Designed and developed a responsive portfolio website to showcase projects, skills, and academic details.
- Implemented using HTML, CSS, and JavaScript with a focus on clean UI and accessibility across devices.

## Education

### **B.Tech. Computer Science and Engineering (AI & ML)**

2023 – 2027

Kalasalingam Academy of Research and Education, Tamil Nadu, India

CGPA: 8.56

### **Intermediate (MPC)**

2021 – 2023

Ignite Junior College, Hyderabad, India

Percentage: 90.4%

## Languages

English, Hindi, Telugu

## Hobbies

Reading, Sports, Problem Solving